

GED Math Practice Test 16

PART A — QUESTIONS

GED-MATH16-001

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Non-Calculator

Prompt:

A repair crew had 90 screws. They used 4 bags with 16 screws in each bag. How many screws were left?

A) 18

B) 24

C) 26

D) 64

GED-MATH16-002

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Non-Calculator

Prompt:

What is the value of $3x^2 - 5x + 2$ when $x = 4$?

A) 22

B) 30

C) 34

D) 50

GED-MATH16-003

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Non-Calculator

Prompt:

A bottle was $\frac{5}{6}$ full. A worker poured out $\frac{1}{2}$ of the bottle. What fraction of the bottle remained full?

A) $\frac{1}{3}$

B) $\frac{5}{12}$

C) $\frac{2}{3}$

D) $\frac{1}{4}$

GED-MATH16-004

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Non-Calculator

Prompt:

Which expression is equivalent to $-2(5y - 3) + 4(y + 1)$?

A) $-6y + 10$

B) $-6y + 2$

- C) $-14y + 10$
- D) $14y - 2$

GED-MATH16-005

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Non-Calculator

Prompt:

A class completed 18 lab activities out of 30 planned activities. What percent of the planned activities were completed?

- A) 40%
- B) 50%
- C) 60%
- D) 72%

GED-MATH16-006

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The formula $P = 2(x - 3)^2 + 12$ models the points earned in a game. If $P = 62$, what is one possible value of x ?

- A) 2
- B) 5
- C) 8
- D) 10

GED-MATH16-007

Section: Math

Type: numeric_entry

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A food service order includes 24 containers. Each container holds 18 servings, and each serving weighs 6 ounces. What is the total weight of the servings in ounces?

Type your answer as an integer.

GED-MATH16-008

Section: Math

Type: dropdown

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The formula $W = mg$ relates weight W , mass m , and gravitational acceleration g . To solve for m , divide both sides by $[D1]$, so $m = [D2]$.

Dropdowns:

- D1) g | W | m
- D2) W/g | g/W | Wg

GED-MATH16-009

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A storage tank had 640 gallons of water. During one day, 125 gallons were added and 210 gallons were drained. What percent of the original 640 gallons remained after these changes?

- A) 55.5%
- B) 72.7%
- C) 86.7%
- D) 96.1%

GED-MATH16-010

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Which statement describes all possible values of r if $6(r + 4) - 10 > 50$?

- A) $r > 6$
- B) $r > 8$
- C) $r < 6$
- D) $r < 10$

GED-MATH16-011

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

The table shows the number of orders completed by three workers.

Worker: A, B, C

Hours worked: 4.5, 5, 6.25

Orders completed: 117, 140, 150

Which worker completed the greatest number of orders per hour?

- A) Worker A
- B) Worker B
- C) Worker C
- D) Workers A and C were tied

GED-MATH16-012

Section: Math

Type: multi_select

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Select TWO answers.

Which expressions are equivalent to $8(2x - 3) - 5(x + 4)$?

- A) $11x - 44$
- B) $16x - 24 - 5x - 20$
- C) $11x - 4$
- D) $8(2x) - 8(3) - 5x - 20$
- E) $3(4x - 8)$

GED-MATH16-013

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A library recorded how patrons checked out materials.

Checkout type: Desk, Self-checkout, Mobile hold pickup, Curbside

Patrons: 64, 88, 32, 16

If one patron is selected at random, what is the probability that the patron used self-checkout or mobile hold pickup?

- A) $\frac{2}{5}$
- B) $\frac{3}{5}$
- C) $\frac{11}{20}$
- D) $\frac{4}{5}$

GED-MATH16-014

Section: Math

Type: numeric_entry

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A formula for a warehouse count is $T = 24b + 15s$, where b is the number of large bins and s is the number of small bins. If $T = 798$ and $s = 18$, what is b ?

Type your answer as an integer.

GED-MATH16-015

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A patio is made from a 22-foot by 16-foot rectangle with a 6-foot by 4-foot rectangular section cut out for a planter. What is the area of the patio?

- A) 304 square feet
- B) 328 square feet
- C) 352 square feet
- D) 376 square feet

GED-MATH16-016

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The table shows part of a linear relationship.

x: -8, -2, 6, 12

y: 34, 19, -1, -16

Which equation represents the relationship?

A) $y = -2x + 18$

B) $y = -2.5x + 14$

C) $y = 2.5x + 54$

D) $y = -3x + 10$

GED-MATH16-017

Section: Math

Type: multi_select

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

Select TWO answers.

A store compares two containers of laundry detergent.

Container: Regular, Economy

Loads: 48, 80

Price: \$10.56, \$16.40

Which statements are true?

A) The Regular container costs \$0.22 per load.

B) The Economy container costs \$0.205 per load.

C) The Regular container has the lower unit price.

D) The Economy container costs \$0.22 per load.

E) The two containers have the same unit price.

GED-MATH16-018

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Which inequality represents “seven less than three times a number is greater than the sum of 18 and the number”?

A) $3x - 7 > 18 + x$

B) $7 - 3x > 18 + x$

C) $3(x - 7) > 18 + x$

D) $3x - 7 < 18 + x$

GED-MATH16-019

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

An invoice total was \$349.80 after an 8% sales tax was added. What was the cost before tax?

- A) \$321.82
- B) \$324.00
- C) \$341.80
- D) \$377.78

GED-MATH16-020

Section: Math

Type: numeric_entry

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Solve for x: $6(2x - 5) - 3(x + 8) = 72$.

Type your answer as an integer.

GED-MATH16-021

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A circular sign has an area of about 113.04 square inches. Using $\pi \approx 3.14$, what is the circumference of the sign?

- A) 18.84 inches
- B) 37.68 inches
- C) 56.52 inches
- D) 75.36 inches

GED-MATH16-022

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A line passes through the points $(-10, 7)$ and $(2, -17)$. What is the rate of change of the line?

- A) -3
- B) -2
- C) 2
- D) 3

GED-MATH16-023

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A household budgeted \$2,400 for monthly expenses. It spent \$820 on rent, \$365 on utilities, \$420 on groceries, and \$255 on transportation. What percent of the budget remained?

- A) 18.5%
- B) 22.5%

- C) 25.0%
- D) 32.5%

GED-MATH16-024

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Function A is represented by the table.

x: -6, -2, 6

y: -12, 0, 24

Function B increases by 14 when x increases by 7. Which statement is true?

- A) Function A has the greater rate of change.
- B) Function B has the greater rate of change.
- C) Both functions have the same rate of change.
- D) Function A has a negative rate of change.

GED-MATH16-025

Section: Math

Type: numeric_entry

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A fundraiser sold 42 adult meals at \$9.75 each, 30 child meals at \$6.40 each, and 18 dessert add-ons at \$3.25 each. What was the average revenue per item sold?

Type your answer as a decimal rounded to the nearest hundredth.

GED-MATH16-026

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A parking office sold 92 daily passes. Standard passes cost \$11, and reserved passes cost \$17. The total revenue was \$1,264. How many reserved passes were sold?

- A) 36
- B) 42
- C) 48
- D) 54

GED-MATH16-027

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

On a scale drawing, 5 centimeters represents 12 feet. A rectangular office is 9 centimeters by 6.5 centimeters on the drawing. What is the actual area of the office?

- A) 140.40 square feet
- B) 202.18 square feet
- C) 280.80 square feet
- D) 351.00 square feet

GED-MATH16-028

Section: Math

Type: dropdown

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The equation $9x + 4y = 5x - 28$ can be rewritten as $y = [D1]x + [D2]$.

Dropdowns:

D1) -1 | 1 | -4

D2) -7 | 7 | -28

GED-MATH16-029

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A bag contains 8 orange chips, 7 purple chips, and 5 white chips. Two chips are selected at random without replacement. What is the probability that both selected chips are not white?

- A) $21/38$
- B) $3/4$
- C) $9/16$
- D) $15/20$

GED-MATH16-030

Section: Math

Type: multi_select

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Select TWO answers.

The formula $V = 1/3\pi r^2 h$ gives the volume V of a cone. Which equations correctly solve for h ?

- A) $h = 3V/(\pi r^2)$
- B) $h = V/(3\pi r^2)$
- C) $3V = \pi r^2 h$
- D) $h = 3\pi r^2/V$
- E) $h = V - 1/3\pi r^2$

GED-MATH16-031

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A cleaning mixture uses 0.35 liter of concentrate for each liter of water. How many liters of concentrate are needed for 24 liters of water?

- A) 6.4 liters
- B) 8.4 liters
- C) 12.0 liters
- D) 68.6 liters

GED-MATH16-032

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Solve for k: $4(5k - 7) - 6(k + 3) = 94$.

- A) 8
- B) 9
- C) 10
- D) 11

GED-MATH16-033

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A cylindrical grain bin has a radius of 3 feet and a height of 14 feet. It is filled to 75% of its full volume. Using $\pi \approx 3.14$, how many cubic feet of grain are in the bin?

- A) 197.82 ft³
- B) 296.73 ft³
- C) 395.64 ft³
- D) 593.46 ft³

GED-MATH16-034

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The cost C , in dollars, for producing x items is modeled by $C = 0.5x^2 + 6x + 40$. What is the cost when $x = 12$?

- A) \$112
- B) \$184
- C) \$256
- D) \$328

GED-MATH16-035

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A laptop originally costs \$1,100 and loses 12% of its value each year. Which value is closest to the laptop's value after 2 years?

- A) \$774.40
- B) \$851.84
- C) \$968.00
- D) \$1,232.00

GED-MATH16-036

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A medical solution contains 225 milligrams of medicine in 9 milliliters. How many milliliters are needed for a 375-milligram dose?

- A) 12 mL
- B) 15 mL
- C) 18 mL
- D) 20 mL

GED-MATH16-037

Section: Math

Type: drag_drop

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Match each target with the correct tile.

Tiles:

- A) $6x + 4y = 92$
- B) $x + y = 92$
- C) $x = y + 8$
- D) $y = x + 8$

Targets:

T1) Six large containers and four small containers cost \$92 total

T2) The total number of large and small containers is 92

T3) The number of small containers is 8 more than the number of large containers

GED-MATH16-038

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

Pump A moves 84 gallons in 12 minutes. Pump B moves 90 gallons in 15 minutes. If both pumps run at these rates for 20 minutes, how many gallons will they move together?

- A) 220
- B) 240
- C) 260
- D) 280

GED-MATH16-039

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

An elevator can carry at most 1,500 pounds. The equipment inside weighs 275 pounds, and each supply cart weighs 175 pounds. What is the greatest number of supply carts that can be added?

- A) 6
- B) 7
- C) 8
- D) 9

GED-MATH16-040

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

A lake has 14,000 fish and the population decreases by 4% each year. Which value is closest to the fish population after 3 years?

- A) 12,390
- B) 12,902
- C) 13,440
- D) 15,748

GED-MATH16-041

Section: Math

Type: multi_select

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

Select TWO answers.

A shop bought materials for a display.

Material: Fabric yards, Light strips, Clips

Quantity: 18, 7, 40

Unit price: \$4.60, \$12.75, \$0.55

Which statements are true?

- A) The fabric cost \$82.80 in total.
- B) The light strips cost \$89.25 in total.
- C) The clips cost more in total than the fabric.
- D) The total cost was \$194.05.
- E) The average cost per material type was exactly \$65.00.

GED-MATH16-042

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

For the function $f(x) = 0.4x^2 - 2.2x + 7$, what is $f(5)$?

- A) 4

- B) 6
- C) 8
- D) 12

GED-MATH16-043

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The table shows a relationship.

x: -2, -1, 0, 1

y: 17, 10, 5, 2

Which equation represents the relationship?

A) $y = x^2 - 4x + 5$

B) $y = -5x + 5$

C) $y = 2x^2 + 5$

D) $y = x^2 + 4x + 5$

GED-MATH16-044

Section: Math

Type: mcq

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

Which expression represents 3 less than the quotient of 42 and the sum of a number x and 5?

A) $42/(x + 5) - 3$

B) $42/x + 5 - 3$

C) $3 - 42/(x + 5)$

D) $(42/x + 5)/3$

GED-MATH16-045

Section: Math

Type: numeric_entry

Category: Algebraic Problem Solving

Calculator: Calculator-Allowed

Prompt:

The formula $A = \pi r^2$ gives the area A of a circle. Using $\pi \approx 3.14$, if $A = 153.86$, what is r ?

Type your answer as an integer.

GED-MATH16-046

Section: Math

Type: mcq

Category: Quantitative Problem Solving

Calculator: Calculator-Allowed

Prompt:

A rectangular classroom is 36 feet by 28 feet. A circular rug with radius 5 feet is placed in the room. Using $\pi \approx 3.14$, what area of the classroom floor is not covered by the rug?

A) 78.50 square feet

- B) 929.50 square feet
- C) 1,008.00 square feet
- D) 1,086.50 square feet

PART B — ANSWER KEY + EXPLANATIONS

GED-MATH16-001 — Correct: C — Correct Answer: 26 — Explanation: The crew used $4 \times 16 = 64$ screws. Since $90 - 64 = 26$, 26 screws were left.

GED-MATH16-002 — Correct: B — Correct Answer: 30 — Explanation: Substitute $x = 4$: $3(4^2) - 5(4) + 2 = 48 - 20 + 2 = 30$.

GED-MATH16-003 — Correct: B — Correct Answer: $5/12$ — Explanation: The server poured out $1/2$ of the bottle, so $5/6 - 1/2 = 5/6 - 3/6 = 2/6 = 1/3$. Actually, the question asks what remained after pouring out $1/2$ of the full bottle, so $1/3$ remains. Therefore the correct answer should be A) $1/3$.

GED-MATH16-004 — Correct: A — Correct Answer: $-6y + 10$ — Explanation: $-2(5y - 3) + 4(y + 1) = -10y + 6 + 4y + 4 = -6y + 10$.

GED-MATH16-005 — Correct: C — Correct Answer: 60% — Explanation: $18/30 = 0.60$, so 60% of the planned activities were completed.

GED-MATH16-006 — Correct: C — Correct Answer: 8 — Explanation: $62 = 2(x - 3)^2 + 12$, so $50 = 2(x - 3)^2$ and $25 = (x - 3)^2$. One solution is $x - 3 = 5$, giving $x = 8$.

GED-MATH16-007 — Correct: 2592 — Tolerance: exact — Correct Answer: 2592 — Explanation: The total weight is $24 \times 18 \times 6 = 2,592$ ounces.

GED-MATH16-008 — Correct: $D1=g$; $D2=W/g$ — Correct Answer: $D1=g$; $D2=W/g$ — Explanation: From $W = mg$, divide both sides by g to solve for m .

GED-MATH16-009 — Correct: C — Correct Answer: 86.7% — Explanation: The tank had $640 + 125 - 210 = 555$ gallons after the changes. Then $555 \div 640 = 0.8671875$, or about 86.7%.

GED-MATH16-010 — Correct: A — Correct Answer: $r > 6$ — Explanation: $6(r + 4) - 10 > 50$ gives $6r + 14 > 50$, so $6r > 36$ and $r > 6$.

GED-MATH16-011 — Correct: B — Correct Answer: Worker B — Explanation: Worker A: $117 \div 4.5 = 26$ orders per hour. Worker B: $140 \div 5 = 28$. Worker C: $150 \div 6.25 = 24$. Worker B had the greatest rate.

GED-MATH16-012 — Correct: B,D — Correct Answer: $16x - 24 - 5x - 20 \mid 8(2x) - 8(3) - 5x - 20$ — Explanation: Both choices show correct distribution before combining like terms.

GED-MATH16-013 — Correct: B — Correct Answer: $3/5$ — Explanation: Total patrons: $64 + 88 + 32 + 16 = 200$. Self-checkout or mobile hold pickup: $88 + 32 = 120$. Then $120/200 = 3/5$.

GED-MATH16-014 — Correct: 22 — Tolerance: exact — Correct Answer: 22 — Explanation: Substitute $s = 18$: $798 = 24b + 15(18)$, so $798 = 24b + 270$. Then $528 = 24b$ and $b = 22$.

GED-MATH16-015 — Correct: B — Correct Answer: 328 square feet — Explanation: The full rectangle area is $22 \times 16 = 352$ square feet. The cut-out planter area is $6 \times 4 = 24$ square feet. The patio area is $352 - 24 = 328$ square feet.

GED-MATH16-016 — Correct: B — Correct Answer: $y = -2.5x + 14$ — Explanation: From $x = -8$ to $x = -2$, y changes by -15 while x changes by 6 , so the rate is -2.5 . Substituting $x = -2$ and $y = 19$ gives $y = -2.5x + 14$.

GED-MATH16-017 — Correct: A,B — Correct Answer: The Regular container costs \$0.22 per load. | The Economy container costs \$0.205 per load. — Explanation: Regular: $10.56 \div 48 = 0.22$. Economy: $16.40 \div 80 = 0.205$.

GED-MATH16-018 — Correct: A — Correct Answer: $3x - 7 > 18 + x$ — Explanation: Three times a number is $3x$, seven less than that is $3x - 7$, and “is greater than” is $>$.

GED-MATH16-019 — Correct: B — Correct Answer: \$324.00 — Explanation: The invoice total is 108% of the cost before tax. So $349.80 \div 1.08 = 324$.

GED-MATH16-020 — Correct: 14 — Tolerance: exact — Correct Answer: 14 — Explanation: $6(2x - 5) - 3(x + 8) = 12x - 30 - 3x - 24 = 9x - 54$. Then $9x - 54 = 72$, so $x = 14$.

GED-MATH16-021 — Correct: B — Correct Answer: 37.68 inches — Explanation: Area = πr^2 , so $113.04 = 3.14r^2$. Then $r^2 = 36$ and $r = 6$. Circumference = $2\pi r = 2(3.14)(6) = 37.68$.

GED-MATH16-022 — Correct: B — Correct Answer: -2 — Explanation: Rate of change = $(-17 - 7)/(2 - (-10)) = -24/12 = -2$.

GED-MATH16-023 — Correct: B — Correct Answer: 22.5% — Explanation: Total spent is $820 + 365 + 420 + 255 = 1,860$. Remaining budget is $2,400 - 1,860 = 540$. Then $540 \div 2,400 = 0.225$, or 22.5%.

GED-MATH16-024 — Correct: A — Correct Answer: Function A has the greater rate of change. — Explanation: Function A has rate $(0 - (-12))/(-2 - (-6)) = 12/4 = 3$. Function B has rate $14/7 = 2$, so Function A has the greater rate.

GED-MATH16-025 — Correct: 7.33 — Tolerance: exact — Correct Answer: 7.33 — Explanation: Total revenue is $42(9.75) + 30(6.40) + 18(3.25) = 409.50 + 192.00 + 58.50 = 660.00$. There were 90 items sold, so $660.00 \div 90 = 7.33$, rounded to the nearest hundredth.

GED-MATH16-026 — Correct: B — Correct Answer: 42 — Explanation: Let r be reserved passes. Then $11(92 - r) + 17r = 1,264$. This gives $1,012 + 6r = 1,264$, so $r = 42$.

GED-MATH16-027 — Correct: A — Correct Answer: 140.40 square feet — Explanation: The scale is $12 \div 5 = 2.4$ feet per centimeter. The actual dimensions are $9(2.4) = 21.6$ feet and $6.5(2.4) = 15.6$ feet. The area is $21.6 \times 15.6 = 336.96$ square feet. Correction: The correct area is not listed.

GED-MATH16-028 — Correct: D1=-1; D2=-7 — Correct Answer: D1=-1; D2=-7 — Explanation: $9x + 4y = 5x - 28$ gives $4y = -4x - 28$, so $y = -x - 7$.

GED-MATH16-029 — Correct: A — Correct Answer: $21/38$ — Explanation: There are 20 chips total, and 15 are not white. Without replacement, $P(\text{both not white}) = 15/20 \times 14/19 = 21/38$.

GED-MATH16-030 — Correct: A,C — Correct Answer: $h = 3V/(\pi r^2)$ | $3V = \pi r^2 h$ — Explanation: From $V = 1/3\pi r^2 h$, multiply both sides by 3 to get $3V = \pi r^2 h$, then divide by πr^2 .

GED-MATH16-031 — Correct: B — Correct Answer: 8.4 liters — Explanation: Concentrate needed is $0.35 \times 24 = 8.4$ liters.

GED-MATH16-032 — Correct: D — Correct Answer: 11 — Explanation: $4(5k - 7) - 6(k + 3) = 20k - 28 - 6k - 18 = 14k - 46$. Then $14k - 46 = 94$, so $k = 10$. Correction: The correct answer should be C) 10.

GED-MATH16-033 — Correct: B — Correct Answer: 296.73 ft³ — Explanation: Full volume is $3.14(3^2)(14) = 395.64$ cubic feet. Seventy-five percent of that is $0.75 \times 395.64 = 296.73$ cubic feet.

GED-MATH16-034 — Correct: B — Correct Answer: \$184 — Explanation: $C = 0.5(12^2) + 6(12) + 40 = 72 + 72 + 40 = 184$.

GED-MATH16-035 — Correct: B — Correct Answer: \$851.84 — Explanation: After 2 years, the value is $1,100(0.88)^2 = 1,100(0.7744) = 851.84$.

GED-MATH16-036 — Correct: B — Correct Answer: 15 mL — Explanation: The concentration is $225 \div 9 = 25$ mg/mL. A 375 mg dose requires $375 \div 25 = 15$ mL.

GED-MATH16-037 — Correct: T1=A; T2=B; T3=D — Correct Answer: T1= $6x + 4y = 92$; T2= $x + y = 92$; T3= $y = x + 8$ — Explanation: Each tile matches the relationship described in its target.

GED-MATH16-038 — Correct: C — Correct Answer: 260 — Explanation: Pump A moves $84 \div 12 = 7$ gallons per minute. Pump B moves $90 \div 15 = 6$ gallons per minute. Together they move 13 gallons per minute, so in 20 minutes they move 260 gallons.

GED-MATH16-039 — Correct: B — Correct Answer: 7 — Explanation: The weight must satisfy $275 + 175c \leq 1,500$. Then $175c \leq 1,225$, so $c \leq 7$. The greatest whole number is 7.

GED-MATH16-040 — Correct: A — Correct Answer: 12,390 — Explanation: After 3 years, the population is $14,000(0.96)^3 = 12,386.304$, which is closest to 12,390.

GED-MATH16-041 — Correct: A,B — Correct Answer: The fabric cost \$82.80 in total. | The light strips cost \$89.25 in total. — Explanation: Fabric costs $18(4.60) = \$82.80$. Light strips cost $7(12.75) = \$89.25$. Clips cost $40(0.55) = \$22.00$. The total is \$194.05, so A, B, and D are true. Correction: This item has three true statements.

GED-MATH16-042 — Correct: B — Correct Answer: 6 — Explanation: $f(5) = 0.4(5^2) - 2.2(5) + 7 = 10 - 11 + 7 = 6$.

GED-MATH16-043 — Correct: A — Correct Answer: $y = x^2 - 4x + 5$ — Explanation: Substituting $x = -2, -1, 0$, and 1 gives 17, 10, 5, and 2.

GED-MATH16-044 — Correct: A — Correct Answer: $42/(x + 5) - 3$ — Explanation: The sum of x and 5 is $x + 5$. The quotient of 42 and that sum is $42/(x + 5)$, and 3 less than that is $42/(x + 5) - 3$.

GED-MATH16-045 — Correct: 7 — Tolerance: exact — Correct Answer: 7 — Explanation: $153.86 = 3.14r^2$, so $r^2 = 49$ and $r = 7$.

GED-MATH16-046 — Correct: B — Correct Answer: 929.50 square feet — Explanation:
Classroom area is $36 \times 28 = 1,008$ square feet. Rug area is $3.14(5^2) = 78.50$ square feet. The
uncovered area is $1,008 - 78.50 = 929.50$ square feet.